

**APPLICATION OF MATRIX ALGEBRA IN PREDICTING
STUDENTS' EXAMINATION SCORES USING MULTIPLE
LINEAR REGRESSION**

A FINAL YEAR PROJECT

**PRESENTED TO THE DEPARTMENT OF
MATHEMATICS**

FACULTY OF PHYSICAL SCIENCES

UNIVERSITY OF CROSS RIVER STATE

**IN PARTIAL FULFILMENT OF THE REQUIREMENTS
FOR THE AWARD OF THE DEGREE OF BACHELOR
OF SCIENCE (B.Sc.) IN MATHEMATICS**

BY

DAVIS PRINCE JACKSON

MATRICULATION NUMBER: 21/MTH/001

SUPERVISOR: Dr. E. E. IYEME

OCTOBER, 2026

DECLARATION

I hereby declare that this project titled “**Application of Matrix Algebra in Predicting Students' Examination Scores Using Multiple Linear Regression**” is my original work and has been carried out by me under the supervision of my appointed supervisor.

The work presented in this project has not been submitted, either wholly or partly, to any other institution for the award of a degree, diploma, certificate, or any other academic qualification. All sources of information and materials used in the course of this study have been duly acknowledged and referenced.

I take full responsibility for the content, analysis, interpretations, and conclusions presented in this project.

Davis Prince Jackson
Student

Date: _____

CERTIFICATION

This is to certify that this project titled “**Application of Matrix Algebra in Predicting Students' Examination Scores Using Multiple Linear Regression**” was carried out by **Davis Prince Jackson** of the Department of Mathematics, Faculty of Physical Sciences, University of Cross River State, under my supervision.

The project has been examined and approved as meeting the requirements for the award of the degree of Bachelor of Science (B.Sc.) in Mathematics.

[Supervisor's Name]

Project Supervisor

Date: _____

[Head of Department's Name]

Head of Department

Date: _____

[External Examiner's Name]

External Examiner

Date: _____

APPROVAL PAGE

This project titled “**Application of Matrix Algebra in Predicting Students' Examination Scores Using Multiple Linear Regression**” has been presented by **Davis Prince Jackson** to the Department of Mathematics, Faculty of Physical Sciences, University of Cross River State, and has been accepted as satisfying the requirements for the award of the degree of Bachelor of Science (B.Sc.) in Mathematics.

[Supervisor's Name]

Project Supervisor

Date: _____

[Head of Department's Name]

Head of Department

Date: _____

[Dean's Name]

Dean, Faculty of [Faculty Name]

Date: _____

External Examiner

Date: _____

DEDICATION

This project is dedicated to God Almighty for His grace, strength, wisdom, and guidance throughout my undergraduate studies.

I also dedicate this work to my family, whose support, encouragement, sacrifices, and prayers have contributed greatly to my academic journey.

Finally, I dedicate this work to everyone who has encouraged me to pursue knowledge, develop my mathematical abilities, and apply mathematics to practical problems in the real world.

ACKNOWLEDGEMENT

I sincerely give thanks to God Almighty for granting me the strength, wisdom, good health, and perseverance required to complete my undergraduate studies and this project successfully.

My profound appreciation goes to my project supervisor, **Dr. E. E. Iyeme**, for the guidance, corrections, constructive criticism, patience, and academic direction provided throughout the development of this research work. The valuable suggestions and observations received during the course of this study contributed significantly to the quality of the project.

I also appreciate the Head of the Department of Mathematics, **Dr. E. E. Iyeme**, and all the lecturers in the Department of Mathematics, University of Cross River State, for the knowledge and academic foundation they provided throughout my undergraduate programme.

I am deeply grateful to my parents, family members, friends, and colleagues for their prayers, encouragement, financial and moral support, and for standing by me throughout my academic journey.

I also appreciate everyone whose contributions, directly or indirectly, made the completion of this project possible.

To all who have contributed to my growth and success, I am sincerely grateful.

ABSTRACT

This study investigated the application of matrix algebra in predicting students' examination scores using multiple linear regression. The study was motivated by the need to demonstrate how mathematical concepts, particularly matrix operations, can be applied to a practical machine learning prediction problem involving students' academic performance. The study used the **Student Habits vs Academic Performance** dataset obtained from Kaggle, containing 1,000 student records and variables describing demographic characteristics, study habits, attendance, sleep, exercise, social media usage, diet, mental health, parental education, internet quality, extracurricular participation, and examination scores.

Data preprocessing involved handling missing values, encoding categorical variables using one-hot encoding, and excluding the student identification variable from the predictors. The resulting data were divided into training and testing sets using an 80:20 split. Multiple linear regression was expressed in matrix form as $y = X\beta + \epsilon$, and the least-squares coefficient vector was obtained from the normal equation

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

The training design matrix contained 800 observations and 19 columns, including the intercept. Model performance on the test set was evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination R^2 . The model produced an MAE of **4.1923**, MSE of **26.5325**, RMSE of **5.1510**, and R^2 of **0.8965**. Five-fold cross-validation produced a mean R^2 of approximately **0.8947**.

The study further examined numerical stability through eigenvalue and condition-number analysis. The condition number of the original design matrix was approximately **1305.59**, while the condition number of $X^T X$ was approximately **1,704,576.31**, illustrating the numerical effect associated with forming the normal equations. After standardization, the condition number of the design matrix was substantially reduced to approximately **1.7131**. Independent calculations using matrix inversion, direct linear solving, and the pseudoinverse produced coefficient differences on the order of 10^{-12} , providing numerical verification of the computed solution. A perturbation analysis was also conducted to examine the sensitivity of predictions to small changes in predictor variables.

The study demonstrates that matrix algebra provides a clear mathematical foundation for multiple linear regression and machine learning prediction, while numerical stability and sensitivity analysis provide additional information beyond prediction accuracy. The findings indicate that study hours, mental health rating, social media usage, exercise frequency, Netflix usage, sleep hours, and attendance were among the variables with the strongest numerical influence on predicted examination scores.

TABLE OF CONTENTS

TITLE PAGE

DECLARATION

CERTIFICATION

DEDICATION

ACKNOWLEDGEMENT

ABSTRACT

TABLE OF CONTENTS

LIST OF TABLES

LIST OF FIGURES

LIST OF SYMBOLS AND ABBREVIATIONS

CHAPTER ONE: INTRODUCTION

- 1.1 Background of the Study
- 1.2 Statement of the Problem
- 1.3 Aim of the Study
- 1.4 Objectives of the Study
- 1.5 Research Questions
- 1.6 Significance of the Study
- 1.7 Scope of the Study
- 1.8 Definition of Key Terms

CHAPTER TWO: LITERATURE REVIEW

- 2.1 Introduction
- 2.2 Concept of Machine Learning
- 2.3 Supervised Learning
- 2.4 Concept of Matrix Algebra
- 2.5 Matrices and Linear System
- 2.6 Matrix Representation of Linear Regression
- 2.7 Multiple Linear Regression
- 2.8 Least-Squares Estimation
- 2.9 Normal Equation
- 2.10 Eigenvalues and Eigenvectors
- 2.11 Matrix Conditioning and Condition Numbers
- 2.12 Numerical Stability
- 2.13 Standardization and Scaling
- 2.14 Perturbation Analysis
- 2.15 Academic Performance Prediction
- 2.16 Review of Related Studies
- 2.17 Empirical Review
- 2.18 Gap in Literature
- 2.19 Summary of Literature Review

CHAPTER THREE: METHODOLOGY

- 3.1 Introduction
- 3.2 Research Design
- 3.3 Source of Data
- 3.4 Variables Used
- 3.5 Data Preprocessing
- 3.6 Encoding Categorical Variables
- 3.7 Construction of the Design Matrix
- 3.8 Formulation of the Regression Model
- 3.9 Least-Squares Estimation
- 3.10 Computation of the Coefficient Vector

- 3.11 Verification of the Regression Solution
- 3.12 Prediction of Examination Scores
- 3.13 Residual Analysis
- 3.14 Evaluation of Predictive Performance
- 3.15 Cross-Validation
- 3.16 Standardized Coefficients
- 3.17 Prediction Sensitivity Analysis
- 3.18 Perturbation Analysis
- 3.19 Eigenvalue Analysis
- 3.20 Condition Number Analysis
- 3.21 Standardization and Conditioning
- 3.22 Coefficient Stability
- 3.23 Prediction Stability
- 3.24 Computational Procedure
- 3.25 Software and Computational Tools

CHAPTER FOUR: RESULTS AND DISCUSSION

- 4.1 Introduction
- 4.2 Construction of the Design Matrix
- 4.3 Estimated Regression Coefficients
- 4.4 Interpretation of Major Regression Coefficients
- 4.5 Model Evaluation
- 4.6 Residual Analysis
- 4.7 Cross-Validation Results
- 4.8 Comparison of Predictor Effects
- 4.9 Prediction Sensitivity
- 4.10 Sensitivity Relative to Natural Variation
- 4.11 Perturbation of the Coefficient Vector
- 4.12 Eigenvalue Analysis
- 4.13 Condition Number Before Standardization
- 4.14 Effect of Standardization
- 4.15 Comparison of Coefficient Stability

- 4.16 Comparison of Solution Methods
- 4.17 Prediction Stability Under Perturbation
- 4.18 Discussion of Major Findings
- 4.19 Summary of Results

CHAPTER FIVE: SUMMARY, CONCLUSION AND RECOMMENDATIONS

- 5.1 Introduction
- 5.2 Summary of the Study
- 5.3 Summary of Major Findings
- 5.4 Conclusion
- 5.5 Contributions of the Study
- 5.6 Recommendations
- 5.7 Limitations of the Study
- 5.8 Suggestions for Further Research
- 5.9 Final Conclusion

References

Appendices

LIST OF TABLES

- Table 3.1: Variables Used in the Study
- Table 4.1: Estimated Regression Coefficients
- Table 4.2: Model Performance
- Table 4.3: Summary of Residuals
- Table 4.4: Five-Fold Cross-Validation Results
- Table 4.5: Standardized Coefficients and Absolute Effects
- Table 4.6: Prediction Sensitivity
- Table 4.7: Numerical Variable Sensitivity
- Table 4.8: Perturbation Results

Table 4.9: Condition Numbers Before Standardization

Table 4.10: Condition Numbers Before and After Standardization

Table 4.11: Differences Between Solution Methods

Table 4.12: Summary of Principal Results

LIST OF ABBREVIATIONS AND SYMBOLS

AI — Artificial Intelligence

CV — Cross-Validation

MAE — Mean Absolute Error

ML — Machine Learning

MSE — Mean Squared Error

OLS — Ordinary Least Squares

RMSE — Root Mean Squared Error

R^2 — Coefficient of Determination

X — Design Matrix

X^T — Transpose of the Design Matrix

$X^T X$ — Cross-product Matrix

X — Vector of Observed Response Values

$\hat{y}$ — Vector of Predicted Response Values

β — Vector of Regression Coefficients

$\hat{\beta}$ — Estimated Regression Coefficients

ε — Error Vector

e — Residual Vector

λ — Eigenvalue

κ — Condition Number